

# Product Architecture

# TalentSphere AI

## Week 1 Deliverables

### AI-Powered Multi-Agent Recruitment Intelligence Platform

Capstone Project 2 - Applied Agentic AI for Engineering Managers

## 1. AI-Powered Recruitment Intelligence Platform

### Problem Statement

Engineering managers and talent teams typically operate across applicant tracking systems, interview feedback documents, spreadsheets, compensation tools, and informal workload knowledge. The result is a delayed and incomplete understanding of sourcing quality, rejection patterns, interviewer capacity, offer outcomes, and pipeline health.

### Proposed solution

TalentSphere AI addresses this fragmentation through a coordinated set of specialized agents rather than a single general-purpose chatbot.

The platform begins by,

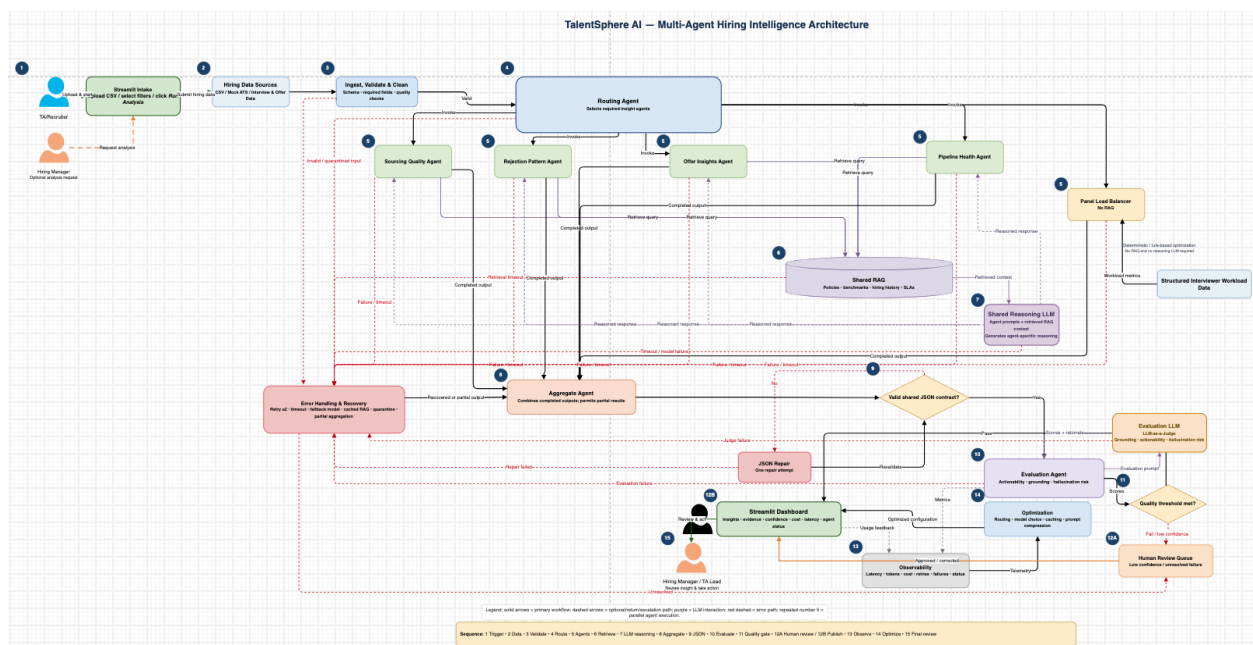
- Ingesting and validating candidate, stage, rejection, offer, source, and interviewer-load data.
- A Routing Agent classifies the request and selects an appropriate model based on complexity, quality requirements, and cost policy.
- It then invokes one or more domain agents: Sourcing Quality, Rejection Pattern, Offer Insights, Pipeline Health, and Panel Load Balancer.
  - Four insight agents retrieve relevant historical context from a shared RAG layer and then reason over both live hiring data and retrieved evidence.
  - The Panel Load Balancer performs deterministic workload analysis without RAG because its recommendations depend primarily on current capacity, calendar load, and role-specific interviewer constraints.

- Completed agent responses flow to an Aggregate Agent, which combines findings without introducing new unsupported claims. A JSON validation layer enforces a shared output contract containing recommendation, evidence, confidence score, cost of insight, and a lower-cost or faster alternative.
- The Evaluation Agent acts as an independent quality gate, scoring actionability, grounding, and hallucination risk against a small golden dataset.
- Observability captures traces, failures, model choices, retrieved context, token usage, latency, and estimated cost.
- Optimization then applies model routing, context limits, prompt compression, retrieval caching, and selective agent execution before results are presented in a Streamlit dashboard.

The system is intentionally positioned as decision support, not automated hiring. It does not rank candidates for employment decisions or replace manager accountability. Its value lies in making operational hiring patterns visible, explainable, and measurable.

## 2. E2E Architecture

The Routing Agent invokes the insight agents. Four insight agents retrieve context from the shared RAG layer and then perform their own reasoning. The Panel Load Balancer operates directly on structured workload data and does not use RAG. Completed agent outputs are passed to the Aggregate Agent, followed by JSON validation and quality evaluation.



## Architecture Principles

1. One clear responsibility per agent. Specialized agents improve prompt focus, testability, and failure isolation.
2. Ground before reasoning. Retrieval occurs before recommendation generation, and retrieved evidence is returned to the same agent that requested it.
3. Aggregate completed outputs only. The Aggregate Agent summarizes and reconciles; it does not replace domain reasoning.
4. Validate and evaluate before display. Schema and quality gates prevent malformed or weak outputs from reaching the dashboard.
5. Route by complexity and risk. Model choice is a policy decision informed by task type, context size, and quality threshold.
6. Human accountability remains explicit. Every insight is advisory, with evidence and uncertainty available for review.

## Key architecture decisions

The Aggregate Agent receives completed recommendations, not raw RAG context. This prevents central aggregation from duplicating specialized reasoning and keeps agent responsibilities clear.

1. Trigger: Manual trigger + http test file used to simulate the trigger.
2. Chunking Strategy:
  - Strategy: Structure-Aware Chunking
  - Why: Hiring data contains tables (e.g., candidate rows). Splitting purely by token count breaks the "Candidate A" row.
3. Embedding & Vector DB:
  - Model: text-embedding-3-small (OpenAI) for high precision.
  - DB: Pinecone (cloud)
  - Retrieval Logic:
    - i. Filter First: The Router passes a filter dictionary (e.g., { "department": "Engineering" }) to the query.
    - ii. Vector Search: Retrieve top\_k = 5 chunks.
    - iii. Rerank: Pass the 5 chunks through a Cohere Rerank model to order them by actual relevance to the specific query
4. Grounding & Prompt
  - System Prompt: "You are a Hiring Intelligence Agent. You must ONLY generate insights based on the text provided in the 'Context' section below. If the context does not explicitly support the answer, you MUST respond with 'Insufficient Evidence'. Cite the specific chunk ID in your evidence field. Do not use outside knowledge about market trends."
5. LLM-as-a-Judge Rubric: The Evaluation Agent checks every output against three criteria derived from Grounding principles:

1. Groundedness (0-1): Is the claim supported by the retrieved evidence?
2. Actionability (0-1): Is the recommendation specific enough to act on (e.g., "Do X" vs. "Think about X")?
3. Confidence Calibration: Does the confidence\_score match the quality of the evidence?

## Core Insight Agents

1. Sourcing Quality Agent
  - Goal: Identify which job boards yield the highest conversion rates.
  - System Prompt: "You are a talent intelligence analyst. Analyze the provided chunks of candidate source data. Recommend whether to increase or decrease spend on specific channels."
2. Rejection Pattern Agent
  - Goal: Cluster rejection reasons (e.g., "Technical Debt" vs "Culture Fit") and flag systemic JD issues.
  - System Prompt: "You are an interviewer coach. Look at the rejection feedback chunks. If a skill appears in >30% of rejections, flag it as a gap. Do not invent reasons not found in the text."
3. Panel Load Balancer
  - Goal: Calculate upcoming interview load and suggest rebalancing.
4. Offer Insights Agent
  - Goal: Identify why candidates decline (Comp vs. Competitor vs. Timing).
5. Pipeline Health Agent
  - Goal: Detect SLA breaches (e.g., "Candidate has been in 'Interview' stage for 14 days").

| Agent             | Primary inputs                                       | Question answered                                        | Output emphasis                                                | Grounding  |
|-------------------|------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------|------------|
| Sourcing Quality  | Candidate source, role, stage progression, hires     | Which channels produce the strongest candidates by role? | Recommendation, source evidence, confidence, cost, alternative | Shared RAG |
| Rejection Pattern | Stage outcomes, rejection reasons, role, feedback    | Where and why are candidates repeatedly rejected?        | Pattern, probable cause, action, supporting records            | Shared RAG |
| Offer Insights    | Offer outcome, compensation, timing, decline reasons | Why are offers declined and what should change?          | Offer action, evidence, confidence, alternative                | Shared RAG |

| Agent               | Primary inputs                                   | Question answered                       | Output emphasis                         | Grounding  |
|---------------------|--------------------------------------------------|-----------------------------------------|-----------------------------------------|------------|
| Pipeline Health     | Stage timestamps, role age, SLA thresholds       | Where are bottlenecks and SLA breaches? | Prioritized bottlenecks and remediation | Shared RAG |
| Panel Load Balancer | Interviewer assignments, capacity, calendar load | Who is overloaded or underutilized?     | Rebalancing plan and workload evidence  | No RAG     |

## Supporting Agents

6. Routing Agent (The "Brain")
  - Pattern: *Routing Pattern*.
  - Logic: A decision tree classifier (LLM-based) that reads the question and determines which agent ID to invoke.
  - Logic Diagram: User Query -> LLM Classifier -> Switch(Node) -> Agent A or Agent B.
7. Optimization Agent
  - Pattern: *Reflection Pattern*.
  - Logic: GPT-4o-mini - The Optimization Agent reviews execution metrics and recommends prompt improvements, retrieval optimization, context-size tuning, or model selection changes based on cost and quality.
8. Evaluation Agent
  - Pattern: *Evaluation Pattern*.
  - Logic: An LLM-as-a-Judge that compares outputs against a Golden Dataset (15 pre-validated scenarios).

## Shared Agent Output Contract

Every insight agent returns the same base structure so the Aggregate Agent, validator, evaluator, and dashboard can process outputs consistently. Agent-specific details may be added under metadata, but the required fields remain stable.

```
{
  "agent_name": "pipeline_health",
  "recommendation": "Prioritize the Senior Backend Engineer role; it has exceeded the 30-day SLA.",
  "evidence": [
    {"metric": "days_open", "value": 44, "source": "mock_ats.csv"},
    {"metric": "onsite_to_offer_rate", "value": 0.12, "source": "historical_hiring_index"}
  ],
  "confidence_score": 0.88,
  "cost_of_insight": {
    "model": "selected-model",
```

```

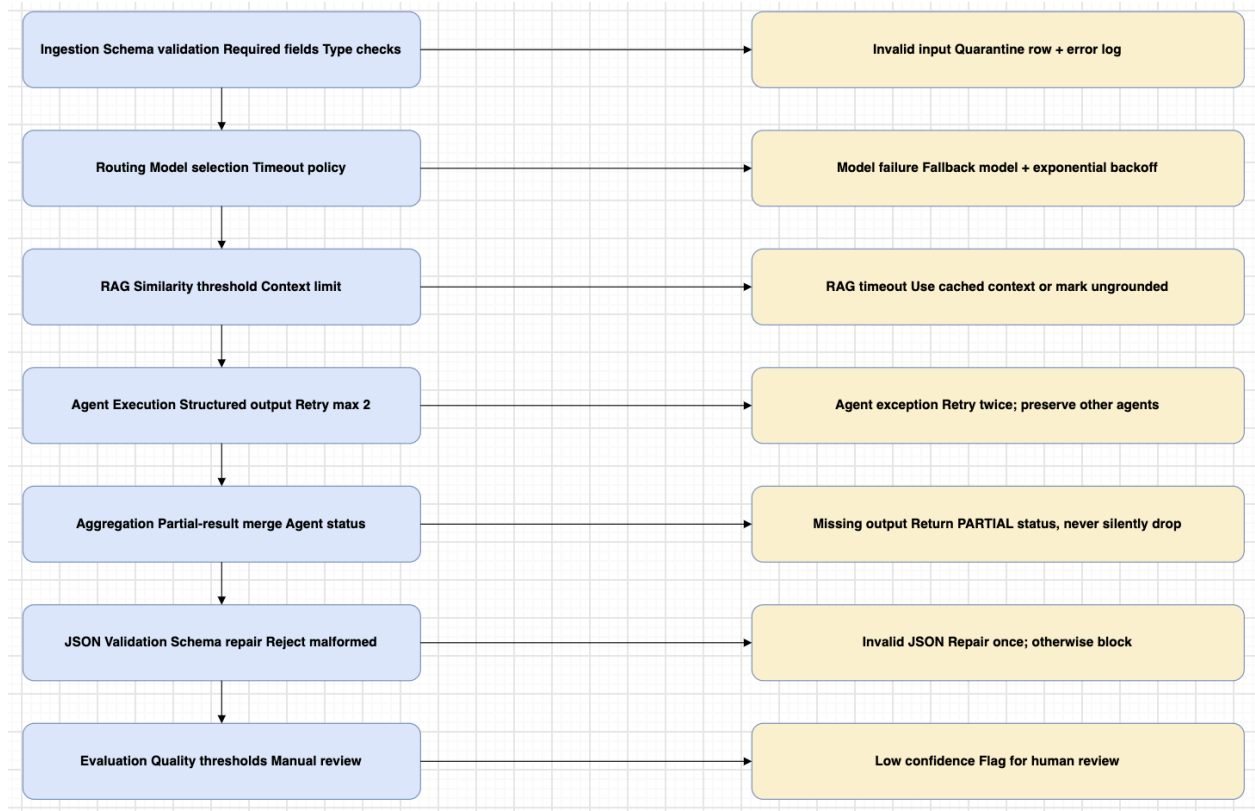
    "input_tokens": 920,
    "output_tokens": 210,
    "latency_ms": 1850,
    "estimated_usd": 0.0041
  },
  "alternative": "Run a deterministic SLA report without the LLM; faster and cheaper, but less explanatory.",
  "status": "SUCCESS",
  "error": null
}

```

| Field            | Requirement                            | Validation rule                                            |
|------------------|----------------------------------------|------------------------------------------------------------|
| recommendation   | Specific and actionable                | Non-empty; must contain an action verb or decision         |
| evidence         | Traceable input or retrieved facts     | At least one source-linked item                            |
| confidence_score | Calibrated score                       | Float from 0.0 to 1.0                                      |
| cost_of_insight  | Model, tokens, latency, cost           | Required numeric telemetry                                 |
| alternative      | Cheaper or faster option and trade-off | Non-empty                                                  |
| status / error   | Operational state                      | SUCCESS, PARTIAL, or FAILED; error required unless SUCCESS |

## 4. Orchestration and Error Handling

A failed agent must not silently terminate the whole run. The orchestrator records status per stage, retries recoverable errors, uses fallbacks where appropriate, and allows partial results to proceed with explicit warnings. This design is essential for the first end-to-end demonstration and supports the evaluation criterion for justified orchestration and handled errors.



| Failure condition             | Recovery behavior                                         | Retry / fallback                         | Required telemetry                            |
|-------------------------------|-----------------------------------------------------------|------------------------------------------|-----------------------------------------------|
| Input validation failure      | Quarantine invalid row; continue valid records            | No retry unless source file is corrected | Validation log and rejected-row count         |
| Routing/model timeout         | Retry with exponential backoff; switch to fallback model  | Maximum two attempts                     | Selected model, fallback reason, elapsed time |
| RAG timeout or low similarity | Use cached context or return ungrounded flag              | One retrieval retry                      | Query, top_k, similarity scores, cache use    |
| Insight agent exception       | Retry agent independently; preserve other outputs         | Maximum two attempts                     | Agent error, retry count, final status        |
| Missing agent result          | Aggregate available results with PARTIAL status           | No hidden omission                       | Missing-agent list and warning                |
| Invalid JSON                  | Attempt deterministic schema repair once; otherwise block | One repair attempt                       | Validation errors and repaired payload        |

| Failure condition    | Recovery behavior                         | Retry / fallback      | Required telemetry                             |
|----------------------|-------------------------------------------|-----------------------|------------------------------------------------|
| Low evaluation score | Do not present as approved recommendation | Human review required | Actionability, grounding, hallucination scores |

## 5. Production Implementation

- Platform: n8n Cloud
- LLM: OpenAI GPT-4o-mini (Responses API)
- Knowledge Base: Google Drive
- Vector Database: Pinecone
- Frontend: Streamlit
- Output: Dashboard-ready JSON
- Status: Successfully tested end-to-end

# Personas

## Personas:

1. TA / Recruiter: Uploads or refreshes hiring data —► Streamlit

TA / Recruiter

|

| Upload CSV / select filters / click Run

▼

Streamlit

|

| Analysis request + uploaded data

▼

Ingest, Validate & Clean

2. Hiring Manager: Requests or reviews analysis —► Streamlit

Hiring Manager

|

| Requests analysis

▼

Streamlit

# Architecture Flow Sequence

# n8n Architecture Flow Sequence

1. Trigger: The TA/Recruiter uploads hiring data, selects filters, and clicks Run Analysis in the Streamlit intake interface.

The Hiring Manager can also submit an optional analysis request, shown as a dashed path.

2. Hiring Data Sources: CSV, mock ATS, interview, and offer data enter the workflow.

3. Ingest, Validate & Clean: The system checks schema, required fields, data types, and data quality.

4. Routing Agent: The Routing Agent determines which specialized agents should execute.

5. Insight Agents: The selected agents execute in parallel:

1. Sourcing Quality
2. Rejection Pattern
3. Offer Insights
4. Pipeline Health
5. Panel Load Balancer

All five carry the same sequence number because they are parallel branches.

6. Pinecone: The four evidence-based agents retrieve policies, benchmarks, hiring history, and SLA information.

- The Panel Load Balancer continues to use structured workload data without RAG.

7. Open AI gpt 4.o mini Reasoning LLM: The LLM reasons over the agent request, calculated metrics, and retrieved evidence. Its reasoned response returns to the calling agent.

8. Aggregate Agent: Completed agent outputs are combined. Partial results can be preserved if one agent fails.

9. JSON Validation: The shared output contract is validated. Invalid output receives one repair attempt.

10. Evaluation Agent: The system evaluates grounding, actionability, evidence support, and hallucination risk using the evaluation LLM.

11. Quality checks: The workflow checks whether the quality threshold is met.

12A. Human Review: Low-confidence, unresolved, or policy-sensitive recommendations are routed to the Human Review Queue.

Approved or corrected results return to Streamlit.

12B. Publish: Results that pass the quality threshold can be published directly to the Streamlit Dashboard.

13. Observability: Latency, token usage, cost, failures, retries, model use, and status are recorded.

- Observability operates throughout the workflow.

14. Optimization: Telemetry is used to improve routing, model selection, caching, and prompt compression for future runs.

15. Final Human Review: The Hiring Manager or TA Lead reviews the insight in Streamlit and takes the appropriate business action.

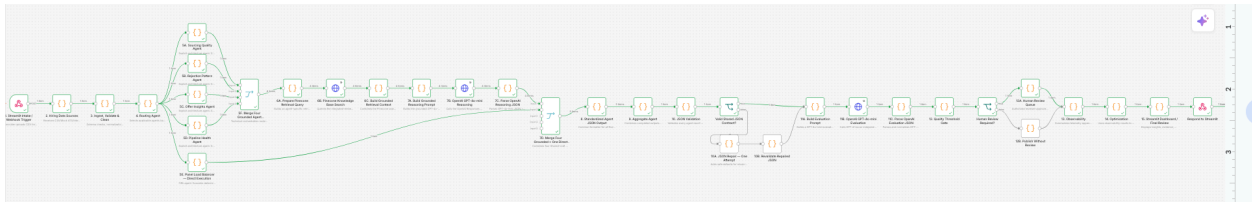
Execution screenshot

## Execution results:

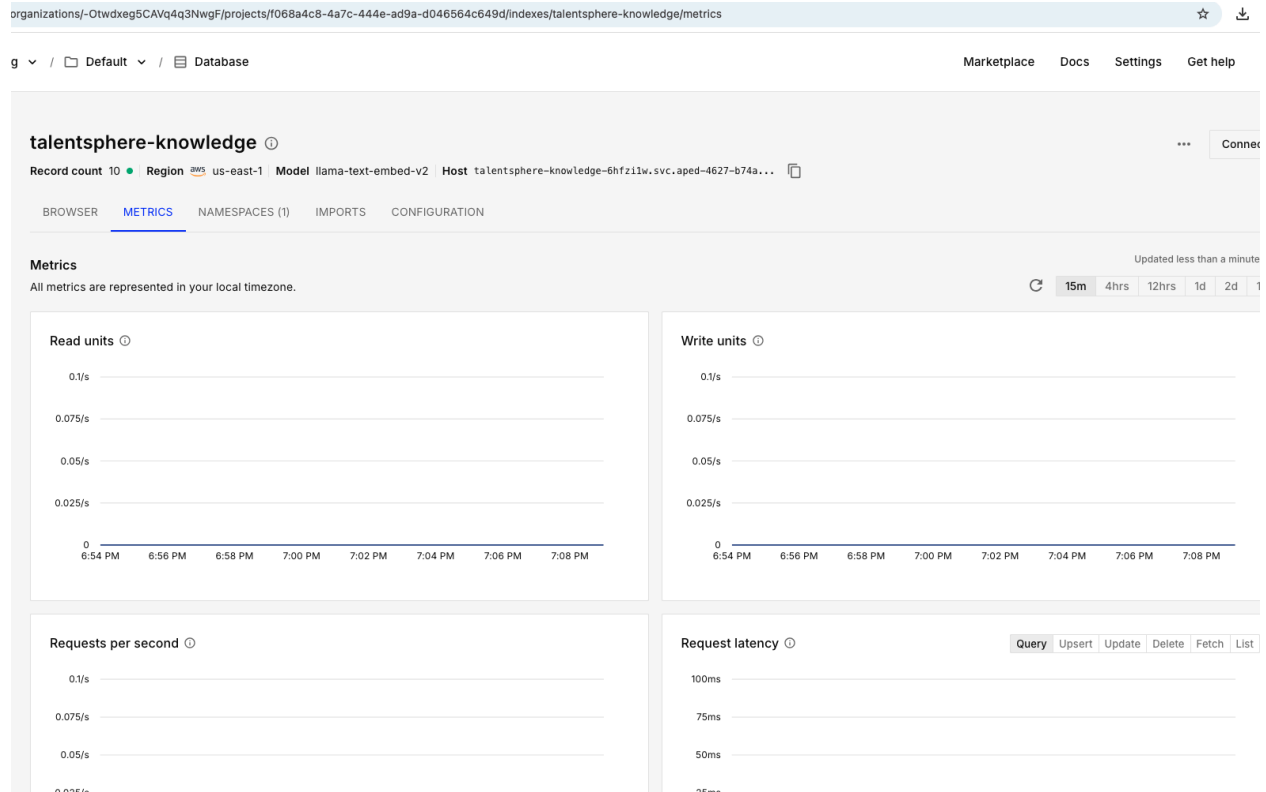
- Successfully orchestrates five specialized agents.
- Retrieves grounded knowledge from Pinecone.
- Generates structured recommendations using GPT-4o-mini.
- Validates and repairs JSON outputs.
- Evaluates responses with an LLM-as-a-Judge.
- Supports human review for low-confidence recommendations.
- Provides observability metrics including latency, token usage, and estimated cost.

### Execution screenshots

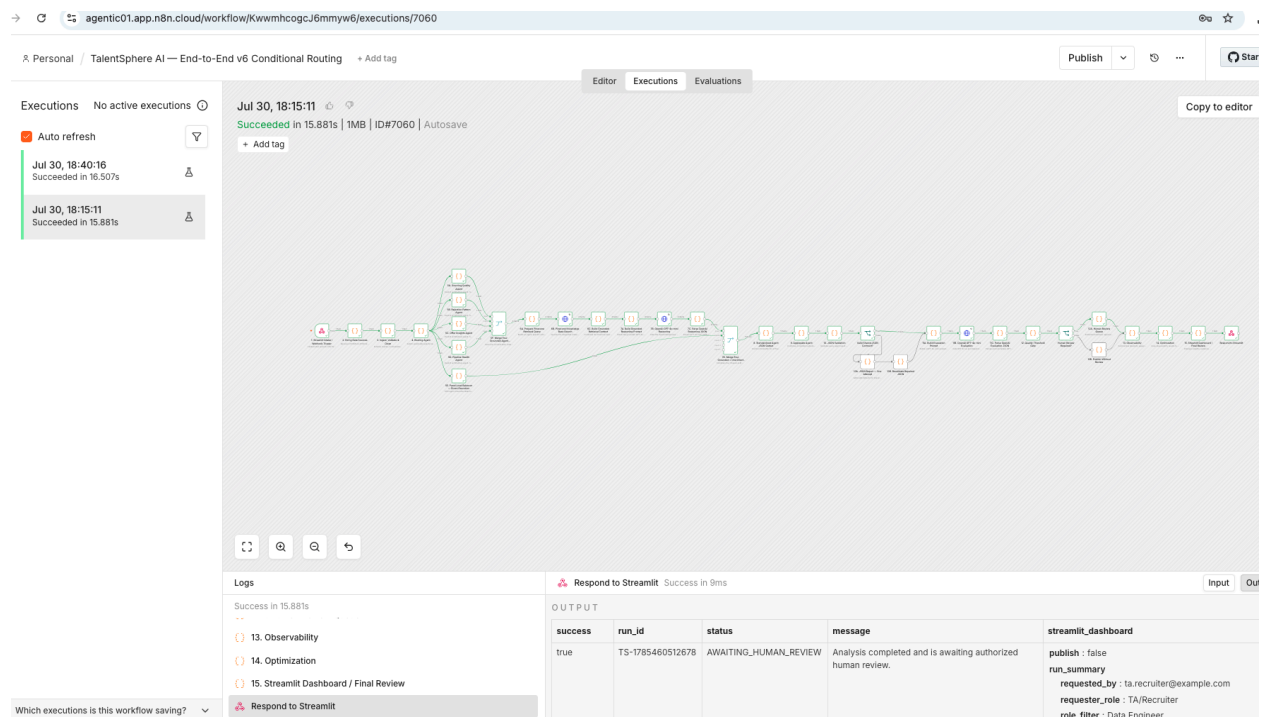
#### 1. n8n workflow

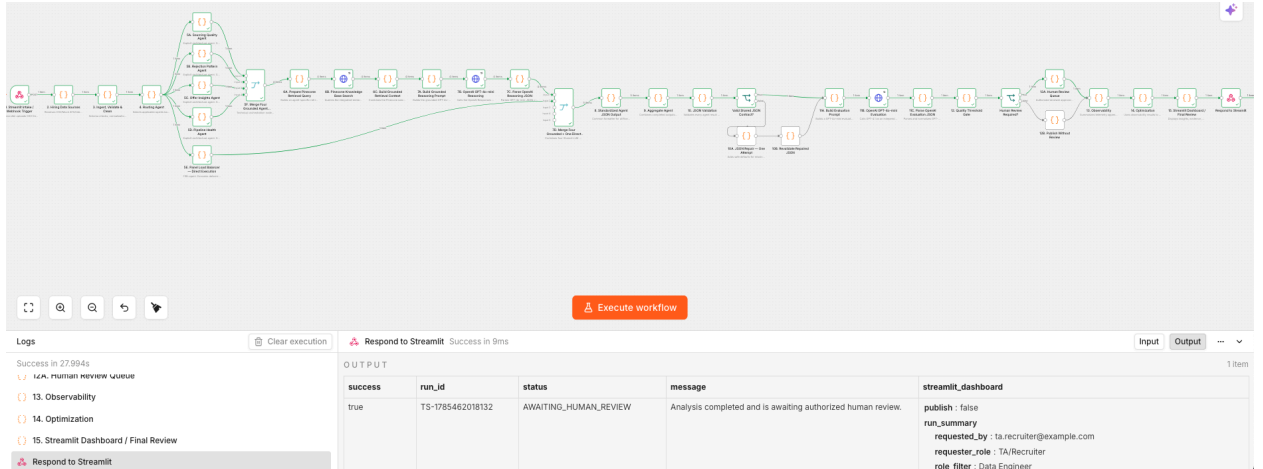


#### 2. Pinecone index



### 3. Workflow execution





[Execute http request to simulate the data load by a TA](#)

```

EXPLORER
  IK_CAPSTONE
    test.http

test.http x
test.http > POST /webhook-test/talentsphere-ai-capstone
Send Request
1 POST https://agentic01.app.n8n.cloud/webhook-test/talentsphere-ai-capstone
2 Content-Type: application/json
3
4 {
5   "requested_by": "ta.recruiter@example.com",
6   "requester_role": "TA/Recruiter",
7   "role_filter": "Data Engineer",
8   "analysis_period": "2026-01-01 to 2026-07-29",
9   "records": [
10     {
11       "candidate_id": "C-1001",
12       "role": "Data Engineer",
13       "department": "Technology",
14       "location": "Seattle, WA",
15       "source": "Employee Referral",
16       "application_date": "2026-01-05",
17       "stage": "Hired",
18       "stage_entry_date": "2026-01-10",
19       "rejection_reason": "",
20       "offer_status": "Accepted",
21       "offer_amount": 142000,
22       "days_to_decision": 3,
23       "interviewer": "Alex",
24       "interview_hours": 2,
25       "agency_fee_pct": 0,
26       "quality_of_hire_score": 4.4
27     },
28     {
29       "candidate_id": "C-1002"

```

#### 4. [Final output JSON](#)

Final output

```
[
{
  "success": true,
  "run_id": "TS-1785462018132",
  "status": "AWAITING_HUMAN_REVIEW",
  "message": "Analysis completed and is awaiting authorized human review.",
  "streamlit_dashboard": {
    "publish": false,
    "run_summary": {
      "requested_by": "ta.recruiter@example.com",
      "requester_role": "TA/Recruiter",
      "role_filter": "Data Engineer",
      "analysis_period": "2026-01-01 to 2026-07-29",
      "valid_records": 48,
      "quarantined_records": 0,
      "selected_agents": [
        "sourcing_quality",
        "rejection_pattern",
        "offer_insights",
        "pipeline_health",
        "panel_load_balancer"
      ],
      "completed_agents": [
        "sourcing_quality",
        "rejection_pattern",
        "offer_insights",
        "pipeline_health",
        "panel_load_balancer"
      ],
      "missing_agents": [],
      "duplicate_agents": [],
      "partial_success": false
    },
    "insights": [
      {
        "agent": "sourcing_quality",
        "status": "SUCCESS",
        "recommendation": "Continue utilizing Employee Referrals and LinkedIn as key sourcing channels while closely monitoring the performance of Agencies and Career Sites.",
        "reasoning": "The analysis indicates that Employee Referrals have the highest quality of hire, with an average quality score of 4.5, compared to other sources. The
```

quarterly sourcing benchmark highlights the importance of maintaining channel diversity and not eliminating channels based on small sample sizes. The average cost of hires is competitive and maintains a favorable cost-to-quality ratio.",

```
"metrics": {
  "total_hires": 20,
  "total_rejected": 18,
  "source_distribution": {
    "Employee Referral": 8,
    "LinkedIn": 5,
    "Agency": 6,
    "Career Site": 6
  },
  "average_offer_amount": 142050,
  "average_days_to_decision": 3.5,
  "average_quality_of_hire_score": 4.4
},
"evidence": [
  {
    "statement": "# Source Diversity Guidance\nDocument ID: POL-DIV-03\nConsider source concentration and diversity before reallocating budget.",
    "source": "POL-DIV-03",
    "title": "POL-DIV-03.md"
  },
  {
    "statement": "# Quarterly Sourcing Benchmark\nDocument ID: KB-SOURCE-2026-04\nEmployee Referral historical hire-rate benchmark is 16%. Do not eliminate a channel solely from a small sample.",
    "source": "KB-SOURCE-2026-04",
    "title": "KB-SOURCE-2026-04.md"
  },
  {
    "statement": "# Agency Fee Policy\nDocument ID: POL-AGENCY-07\nStandard agency fee is 18% of first-year base salary. Evaluate cost, quality, and speed together.",
    "source": "POL-AGENCY-07",
    "title": "POL-AGENCY-07.md"
  }
],
"confidence_score": 0.8,
"cost_of_insight": {
  "input_tokens": 5210,
  "cached_input_tokens": 0,
```

```
    "output_tokens": 259,
    "total_tokens": 5469,
    "latency_ms": 7983,
    "estimated_cost_usd": 0.000937
  },
  "alternative": "Collect an additional reporting period and rerun before
making a major policy change.",
  "limitations": [
    "The sample size for some sources, like Agencies, is small, potentially
skewing performance metrics."
  ],
  "requires_human_review": false,
  "human_review_reason": ""
},
{
  "agent": "rejection_pattern",
  "status": "SUCCESS",
  "recommendation": "Implement a standardized rejection reason framework to
reduce vague and inconsistent rejections, especially regarding technical gaps and
remote-work preferences.",
  "reasoning": "The evidence shows a high frequency of rejections due to
'Technical gap' (10 instances) and 'Remote-work preference' (7 instances), indicating
potential inconsistencies in evaluating candidates. Implementing a standardized
rejection taxonomy aligns with policy requirements to document consistent and
job-related reasons to enhance fairness and clarity.",
  "metrics": {
    "total_candidates": 48,
    "hired_candidates": 25,
    "rejected_candidates": 18,
    "offer_declined": 10,
    "technical_gap_rejections": 10,
    "remote_work_preference_rejections": 7,
    "days_to_decision_mean": 4.27
  },
  "evidence": [
    {
      "statement": "# Approved Rejection Taxonomy\nDocument ID: POL-REJ-01\nUse
standardized, job-related rejection reasons and review vague or inconsistent
categories.",
      "source": "POL-REJ-01",
      "title": "POL-REJ-01.md"
    }
  ],
}
```

```
{
  "statement": "# Fair Hiring Review Policy\nDocument ID: POL-FAIR-02\nPotentially biased patterns require HR or compliance review. Protected-class inference is prohibited.",
  "source": "POL-FAIR-02",
  "title": "POL-FAIR-02.md"
},
{
  "confidence_score": 0.8,
  "cost_of_insight": {
    "input_tokens": 5129,
    "cached_input_tokens": 0,
    "output_tokens": 224,
    "total_tokens": 5353,
    "latency_ms": 7983,
    "estimated_cost_usd": 0.000904
  },
  "alternative": "Collect an additional reporting period and rerun before making a major policy change.",
  "limitations": [
    "The analysis does not include deeper qualitative data from interviewer notes or candidate experiences."
  ],
  "requires_human_review": false,
  "human_review_reason": ""
},
{
  "agent": "offer_insights",
  "status": "SUCCESS",
  "recommendation": "Ensure prompt human review for offers that fall below the established compensation band.",
  "reasoning": "With the approved compensation band for Data Engineers in Seattle ranging from $135,000 to $155,000, offers below this threshold could lead to declined offers and ultimately impact hiring goals. Monitoring and adjusting offers in line with this band could help improve acceptance rates.",
  "metrics": {
    "total_candidates": 48,
    "hired_candidates": 18,
    "accepted_offers": 13,
    "declined_offers": 8,
    "rejected_candidates": 24,
    "average_days_to_decision": 3.5,
  }
}
```

```
    "average_offer_amount": 143000
  },
  "evidence": [
    {
      "statement": "# Offer Governance\nDocument ID: POL-OFFER-04\nAI may recommend review but cannot approve compensation or policy exceptions.",
      "source": "POL-OFFER-04",
      "title": "POL-OFFER-04.md"
    },
    {
      "statement": "# Offer Approval SLA\nDocument ID: OFFER-SLA-02\nFinal offer approval target is three business days; escalate beyond five.",
      "source": "OFFER-SLA-02",
      "title": "OFFER-SLA-02.md"
    },
    {
      "statement": "# Data Engineer Compensation Band\nDocument ID: COMP-BAND-DE-2026\nSeattle approved base salary range is $135,000 to $155,000.",
      "source": "COMP-BAND-DE-2026",
      "title": "COMP-BAND-DE-2026.md"
    }
  ],
  "confidence_score": 0.8,
  "cost_of_insight": {
    "input_tokens": 5190,
    "cached_input_tokens": 0,
    "output_tokens": 219,
    "total_tokens": 5409,
    "latency_ms": 7982,
    "estimated_cost_usd": 0.00091
  },
  "alternative": "Collect an additional reporting period and rerun before making a major policy change.",
  "limitations": [
    "The analysis is based on the available hiring records, which may not fully represent future trends or candidate preferences."
  ],
  "requires_human_review": true,
  "human_review_reason": "Compensation recommendations need authorized human approval."
},
{
```

```
    "agent": "pipeline_health",
    "status": "SUCCESS",
    "recommendation": "Increase effort on candidate engagement during the offer
stage to reduce decline rates and improve conversion.",
    "reasoning": "The data shows a significant number of rejections due to
remote-work preferences at the offer stage, indicating that candidates may not be
aligned with the offered terms. Streamlining this stage and enhancing communication
about the role's specifics can help address candidate concerns and reduce declines,
improving overall hiring effectiveness.",
    "metrics": {
        "total_candidates": 48,
        "hired_candidates": 22,
        "rejected_candidates": 26,
        "offer_declined": 10,
        "interview_stage": 4,
        "average_quality_of_hire": 4.38,
        "average_days_to_decision": 4.04,
        "bottleneck_stage": "Offer"
    },
    "evidence": [
        {
            "statement": "# Pipeline Conversion Benchmarks\nDocument ID:
PIPE-BENCH-06\nAnalyze conversion by role, source, stage, and time window.",
            "source": "PIPE-BENCH-06",
            "title": "PIPE-BENCH-06.md"
        },
        {
            "statement": "# Interview Scheduling SLA\nDocument ID:
SLA-INTERVIEW-03\nReview interview-stage candidates after seven inactive days and
offer-stage candidates after five days.",
            "source": "SLA-INTERVIEW-03",
            "title": "SLA-INTERVIEW-03.md"
        }
    ],
    "confidence_score": 0.8,
    "cost_of_insight": {
        "input_tokens": 5131,
        "cached_input_tokens": 0,
        "output_tokens": 225,
        "total_tokens": 5356,
        "latency_ms": 7982,
        "estimated_cost_usd": 0.000905
    }
}
```

```
    },
    "alternative": "Collect an additional reporting period and rerun before
making a major policy change.",
    "limitations": [
        "The analysis is based exclusively on historical data without considering
current job market conditions."
    ],
    "requires_human_review": false,
    "human_review_reason": ""
},
{
    "agent": "panel_load_balancer",
    "status": "SUCCESS",
    "recommendation": "Review moving one suitable interview block from Jordan to
Morgan, subject to skill and availability checks.",
    "reasoning": "Calculated directly from structured interviewer workload data.
No RAG or reasoning LLM was used.",
    "metrics": [
        {
            "interviewer": "Jordan",
            "hours": 48
        },
        {
            "interviewer": "Priya",
            "hours": 36
        },
        {
            "interviewer": "Alex",
            "hours": 24
        },
        {
            "interviewer": "Morgan",
            "hours": 12
        }
    ],
    "evidence": [
        {
            "statement": "Workload hours were calculated from the uploaded
interviewer data.",
            "source": "Hiring dataset",
            "title": "Uploaded panel workload data"
        }
    ]
}
```

```
    ],
    "confidence_score": 0.95,
    "cost_of_insight": {
      "input_tokens": 0,
      "output_tokens": 0,
      "latency_ms": 80,
      "estimated_cost_usd": 0
    },
    "alternative": "Retain the current schedule and review workload again after the next interview cycle.",
    "limitations": [
      "Skill fit and actual calendar availability must be verified before reassignment."
    ],
    "requires_human_review": true,
    "human_review_reason": "Panel assignments affect people and schedules and require an authorized coordinator or manager."
  }
],
"evaluation": {
  "evaluator": "gpt-4o-mini-2024-07-18",
  "response_id": "resp_06aedic0efb597936016a6bfd0bbf0481a0a8df831a6f811a7f",
  "prompt_version": "llm-judge-v2",
  "status": "PASS_WITH_HUMAN_REVIEW",
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      "actionability_score": 0.8,
      "hallucination_risk": 0.1,
      "safety_score": 0.7,
      "decision": "PASS",
      "rationale": "The claims are well-supported by multiple evidence sources detailing the performance and rationale behind sourcing channels."
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      "rationale": "Recommendations are solidly backed by evidence that highlights rejection reasons and policy requirements."
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      "hallucination_risk": 0.2,
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      "rationale": "The recommendation aligns with compensation policies, but requires human review to ensure compliance with nuanced hiring practices."
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      "actionability_score": 0.85,
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      "actionability_score": 0.8,
      "hallucination_risk": 0,
      "safety_score": 0.6,
      "decision": "PASS_WITH_HUMAN_REVIEW",
      "rationale": "Findings are based on data; however, skill fit and availability need human verification before action."
    }
  ],
  "overall_summary": "All agents provided actionable recommendations well-supported by evidence, with most decisions passing. Offer insights and panel load balancing require human review due to the complexity of implementation.",
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    "evaluation_agent",
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    "Keep deterministic metric calculations outside the LLM.",
    "Execute independent insight agents in parallel.",
    "Use a lower-cost reasoning model for low-risk agents and a stronger model
for Offer Insights.",
```

```
        "Compress retrieved context when token usage exceeds the configured  
threshold."  
    ]  
}  
}  
}  
]  
]
```

http test trigger

POST https://agentic01.app.n8n.cloud/webhook-test/talentsphere-ai-capstone

Content-Type: application/json

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      "stage_entry_date": "2026-01-14",
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  "offer_amount": 146000,
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  "interviewer": "Alex",
  "interview_hours": 2,
  "agency_fee_pct": 0,
  "quality_of_hire_score": 4.7
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  "role": "Data Engineer",
  "department": "Technology",
  "location": "Seattle, WA",
  "source": "LinkedIn",
  "application_date": "2026-04-02",
  "stage": "Hired",
  "stage_entry_date": "2026-04-12",
  "rejection_reason": "",
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  "offer_amount": 140000,
  "days_to_decision": 3,
  "interviewer": "Priya",
  "interview_hours": 3,
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    "quality_of_hire_score": 4.1
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    "source": "Agency",
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    "stage_entry_date": "2026-04-16",
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    "offer_status": "",
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    "days_to_decision": "",
    "interviewer": "Jordan",
    "interview_hours": 4,
    "agency_fee_pct": 18,
    "quality_of_hire_score": ""
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  {
    "candidate_id": "C-1032",
    "role": "Data Engineer",
    "department": "Technology",
    "location": "Seattle, WA",
    "source": "Career Site",
    "application_date": "2026-04-08",
    "stage": "Offer",
    "stage_entry_date": "2026-04-20",
    "rejection_reason": "Remote-work preference",
    "offer_status": "Declined",
    "offer_amount": 134000,
    "days_to_decision": 6,
    "interviewer": "Morgan",
    "interview_hours": 1,
    "agency_fee_pct": 0,
    "quality_of_hire_score": ""
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    "role": "Data Engineer",
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    "location": "Seattle, WA",
    "source": "Employee Referral",
    "application_date": "2026-04-11",
    "stage": "Hired",
    "stage_entry_date": "2026-04-24",
    "rejection_reason": "",
    "offer_status": "Accepted",
    "offer_amount": 144000,
    "days_to_decision": 3,
    "interviewer": "Alex",
    "interview_hours": 2,
    "agency_fee_pct": 0,
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    "department": "Technology",
    "location": "Seattle, WA",
    "source": "LinkedIn",
    "application_date": "2026-04-14",
    "stage": "Rejected",
    "stage_entry_date": "2026-04-28",
    "rejection_reason": "Technical gap",
    "offer_status": "",
    "offer_amount": "",
    "days_to_decision": "",
    "interviewer": "Priya",
    "interview_hours": 3,
    "agency_fee_pct": 0,
    "quality_of_hire_score": ""
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  {
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    "role": "Data Engineer",
    "department": "Technology",
    "location": "Seattle, WA",
    "source": "Agency",
    "application_date": "2026-04-17",
    "stage": "Hired",
    "stage_entry_date": "2026-05-02",
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    "days_to_decision": 3,
    "interviewer": "Jordan",
    "interview_hours": 4,
    "agency_fee_pct": 18,
    "quality_of_hire_score": 4.2
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    "department": "Technology",
    "location": "Seattle, WA",
    "source": "Career Site",
    "application_date": "2026-04-20",
    "stage": "Offer",
    "stage_entry_date": "2026-04-24",
    "rejection_reason": "Remote-work preference",
    "offer_status": "Declined",
    "offer_amount": 134000,
    "days_to_decision": 6,
    "interviewer": "Morgan",
    "interview_hours": 1,
    "agency_fee_pct": 0,
    "quality_of_hire_score": ""
  },
  {
    "candidate_id": "C-1037",
    "role": "Data Engineer",
    "department": "Technology",
    "location": "Seattle, WA",
    "source": "Employee Referral",
    "application_date": "2026-04-23",
    "stage": "Hired",
    "stage_entry_date": "2026-04-28",
    "rejection_reason": "",
    "offer_status": "Accepted",
    "offer_amount": 142000,
    "days_to_decision": 3,
    "interviewer": "Alex",
    "interview_hours": 2,
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    "source": "LinkedIn",
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    "stage_entry_date": "2026-05-02",
    "rejection_reason": "Technical gap",
    "offer_status": "",
    "offer_amount": "",
    "days_to_decision": "",
    "interviewer": "Priya",
    "interview_hours": 3,
    "agency_fee_pct": 0,
    "quality_of_hire_score": ""
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  {
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    "role": "Data Engineer",
    "department": "Technology",
    "location": "Seattle, WA",
    "source": "Agency",
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    "stage": "Rejected",
    "stage_entry_date": "2026-05-06",
    "rejection_reason": "Technical gap",
    "offer_status": "",
    "offer_amount": "",
    "days_to_decision": "",
    "interviewer": "Jordan",
    "interview_hours": 4,
    "agency_fee_pct": 18,
    "quality_of_hire_score": ""
  },
  {
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    "role": "Data Engineer",
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    "source": "Career Site",
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    "stage": "Hired",
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    "offer_amount": 140000,
    "days_to_decision": 4,
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    "interview_hours": 1,
    "agency_fee_pct": 0,
    "quality_of_hire_score": 4.0
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    "department": "Technology",
    "location": "Seattle, WA",
    "source": "Employee Referral",
    "application_date": "2026-05-05",
    "stage": "Hired",
    "stage_entry_date": "2026-05-14",
    "rejection_reason": "",
    "offer_status": "Accepted",
    "offer_amount": 146000,
    "days_to_decision": 3,
    "interviewer": "Alex",
    "interview_hours": 2,
    "agency_fee_pct": 0,
    "quality_of_hire_score": 4.4
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  {
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    "role": "Data Engineer",
    "department": "Technology",
    "location": "Seattle, WA",
    "source": "LinkedIn",
    "application_date": "2026-05-08",
    "stage": "Hired",
    "stage_entry_date": "2026-05-18",
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    "rejection_reason": "",
    "offer_status": "Accepted",
    "offer_amount": 140000,
    "days_to_decision": 3,
    "interviewer": "Priya",
    "interview_hours": 3,
    "agency_fee_pct": 0,
    "quality_of_hire_score": 4.1
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    "source": "Agency",
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    "stage_entry_date": "2026-05-22",
    "rejection_reason": "Technical gap",
    "offer_status": "",
    "offer_amount": "",
    "days_to_decision": "",
    "interviewer": "Jordan",
    "interview_hours": 4,
    "agency_fee_pct": 18,
    "quality_of_hire_score": ""
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  {
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    "role": "Data Engineer",
    "department": "Technology",
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    "source": "Career Site",
    "application_date": "2026-05-14",
    "stage": "Offer",
    "stage_entry_date": "2026-05-26",
    "rejection_reason": "Remote-work preference",
    "offer_status": "Declined",
    "offer_amount": 134000,
    "days_to_decision": 6,
    "interviewer": "Morgan",
    "interview_hours": 1,
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    "source": "Employee Referral",
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    "stage_entry_date": "2026-05-30",
    "rejection_reason": "",
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    "offer_amount": "",
    "days_to_decision": "",
    "interviewer": "Alex",
    "interview_hours": 2,
    "agency_fee_pct": 0,
    "quality_of_hire_score": ""
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  {
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    "role": "Data Engineer",
    "department": "Technology",
    "location": "Seattle, WA",
    "source": "LinkedIn",
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    "stage": "Rejected",
    "stage_entry_date": "2026-06-03",
    "rejection_reason": "Technical gap",
    "offer_status": "",
    "offer_amount": "",
    "days_to_decision": "",
    "interviewer": "Priya",
    "interview_hours": 3,
    "agency_fee_pct": 0,
    "quality_of_hire_score": ""
  },
  {
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    "role": "Data Engineer",
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    "offer_amount": "",
    "days_to_decision": "",
    "interviewer": "Jordan",
    "interview_hours": 4,
    "agency_fee_pct": 18,
    "quality_of_hire_score": ""
  },
  {
    "candidate_id": "C-1048",
    "role": "Data Engineer",
    "department": "Technology",
    "location": "Seattle, WA",
    "source": "Career Site",
    "application_date": "2026-05-26",
    "stage": "Offer",
    "stage_entry_date": "2026-05-30",
    "rejection_reason": "Remote-work preference",
    "offer_status": "Declined",
    "offer_amount": 134000,
    "days_to_decision": 6,
    "interviewer": "Morgan",
    "interview_hours": 1,
    "agency_fee_pct": 0,
    "quality_of_hire_score": ""
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}
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